Comprehension Questions: Function B1

Direction A: Branch Conditions

A1. Under what condition does the function assign self.url directly and return before calling parse_url?

A. The URL is empty.

B. The URL contains ":" and does not start with "http".

C. The URL contains a non-ASCII hostname.

D. The URL has no query parameters.

A2. How does the function handle a LocationParseError raised by parse_url(url)?

A. Return an empty string.

B. Ignore the error and continue constructing the URL.

C. Convert it to an InvalidURL exception and raise it.

D. Assign the original URL directly to self.url.

A3. What happens if the URL has no scheme?

A. Automatically add http://.

B. Raise MissingSchema.

C. Raise InvalidURL("No host supplied").

D. Return the original URL.

Direction B: Path Outcomes and Differences

B1. What happens if the hostname contains non-ASCII characters?

A. The URL is rejected immediately.

B. The function attempts encoding with _get_idna_encoded_host; if encoding fails, it raises InvalidURL.

C. The non-ASCII characters are removed.

D. host is set to an empty string.

B2. If the URL already has a query and the encoded params are non-empty, how are they combined?

A. The new parameters overwrite the old query.

B. The new parameters are ignored.

C. Use query + "&" + enc_params.

D. Move the query into the fragment.

B3. What does the function do if the parsed path is empty?

A. Leave it as an empty string.

B. Set it to "/".

C. Raise an exception.

D. Use the host as the path.

Direction C: Process Convergence and Local Evidence

C1. Which location is the most suitable boundary between “input validation” and “URL component assembly”?

A. url = url.lstrip()

B. if not scheme:

C. netloc = auth or ""

D. self.url = url

C2. If auth='user', host='example.com', and port=8080, what is the assembled netloc?

A. example.com@user:8080

B. user@example.com:8080

C. user:8080@example.com

D. http://user@example.com

C3. Which step provides the value ultimately assigned to self.url?

A. Use the return value of parse_url directly.

B. Reassemble the URL with urlunparse and then normalize it with requote_uri.

C. Concatenate only host and path.

D. Encode only params.

Overall Question

T1. Which option best summarizes the function?

A. Unconditionally convert any string into an HTTP URL.

B. Preprocess, parse, validate, and reassemble URL components, merge query parameters, and write the result back to the request object.

C. Handle query-parameter encoding only.

D. Handle internationalized-domain encoding only.